

Dynamic Guideline: WHO-Based COVID-19 - Diagnosis and Treatment and Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Therapeutics and COVID-19: Living Guideline
- **Publication Date:** The WHO living guideline for COVID-19 therapeutics has been iteratively updated, with the most recent major update published on 11 August 2025 (18th update). The living guideline process continues to incorporate new evidence on an ongoing basis.
- **Target Population:** Patients with confirmed or suspected COVID-19 across all severity spectrums — mild, moderate, severe, and critical illness — including special populations (immunocompromised, pregnant individuals, children, and the elderly).
- **Scope and Objectives:**
 - Provide global, evidence-based recommendations on pharmacological therapeutics for COVID-19.
 - Cover antiviral agents, immunomodulators, neutralizing monoclonal antibodies, anticoagulants, and adjunctive therapies.
 - Guide clinicians, policymakers, and health systems in resource-limited and resource-rich settings.
 - Employ a living guideline methodology to rapidly integrate emerging trial data.

Five Key WHO Guidelines Synthesized:

#	Guideline Title	Date	Scope
1	Therapeutics and COVID-19: Living Guideline (18th update)	11 August 2025	Pharmacological treatment across severity spectrums
2	Clinical Management of COVID-19: Living Guideline	15 January 2025 (latest update)	Comprehensive clinical care pathways including oxygen, ventilation, rehabilitation
3	COVID-19 Vaccine: WHO Position Papers and SAGE Recommendations	Updated through 2025	Vaccination strategy, boosters, variant-adapted vaccines
4	Infection Prevention and Control During Health Care When COVID-19 Is Suspected or Confirmed: Living Guideline	Updated 2024–2025	IPC measures in healthcare settings
5	Meaningful Youth Engagement in a WHO Guideline Development Process: Case Study of WHO COVID-19 Guideline	29 April 2026	Methodological case study on inclusive guideline development processes applied to COVID-19 guidelines

1.2 Key Recommendations

1.2.1 Antiviral Therapies

Recommendation	GRADE Evidence Quality	Strength
Nirmatrelvir-ritonavir (Paxlovid) is recommended for patients with non-severe COVID-19 at highest risk of hospitalization (e.g., elderly, immunocompromised, unvaccinated)	Moderate	Strong
Remdesivir (3-day course) is suggested for patients with non-severe COVID-19 at high risk of hospitalization	Moderate	Conditional
Remdesivir is recommended for patients with severe COVID-19	Moderate	Conditional
Molnupiravir is suggested only when nirmatrelvir-ritonavir and remdesivir are unavailable, for high-risk non-severe patients	Low	Conditional (against use in most contexts)

1.2.2 Immunomodulators

Recommendation	GRADE Evidence Quality	Strength
Systemic corticosteroids (dexamethasone 6 mg/day for up to 10 days) are recommended for patients with severe and critical COVID-19	Moderate	Strong
Systemic corticosteroids should NOT be used in non-severe COVID-19	Moderate	Strong (against)
IL-6 receptor blockers (tocilizumab, sarilumab) are recommended for patients with severe or critical COVID-19, in combination with corticosteroids	Moderate	Strong
Baricitinib (JAK inhibitor) is recommended for patients with severe or critical COVID-19, as an alternative to IL-6 receptor blockers, in combination with corticosteroids	Moderate	Strong

1.2.3 Anticoagulation

Recommendation	GRADE Evidence Quality	Strength
Prophylactic-dose anticoagulation (e.g., LMWH) is recommended for hospitalized patients with COVID-19	Moderate	Strong
Therapeutic-dose anticoagulation with heparin is suggested for non-critically ill hospitalized patients with COVID-19 who do not have a high bleeding risk	Moderate	Conditional
Therapeutic-dose anticoagulation is NOT suggested for critically ill patients with COVID-19	Moderate	Conditional (against)

1.2.4 Therapies NOT Recommended

Recommendation	GRADE Evidence Quality	Strength
Hydroxychloroquine is NOT recommended for COVID-19 (any severity)	High	Strong (against)
Lopinavir-ritonavir is NOT recommended for COVID-19 (any severity)	Moderate	Strong (against)
Ivermectin is NOT recommended for COVID-19 outside clinical trials	Low	Conditional (against)
Convalescent plasma is NOT recommended for non-severe COVID-19; conditional recommendation against use in severe/critical COVID-19	Moderate	Strong (against) for non-severe; Conditional (against) for severe
Fluvoxamine is NOT recommended for COVID-19	Low	Conditional (against)
Colchicine is NOT recommended for COVID-19	Low	Conditional (against)
Sotrovimab, bebtelovimab, and other anti-SARS-CoV-2 monoclonal antibodies are NOT currently recommended due to loss of neutralizing activity against circulating Omicron-lineage variants	Very Low (for current variants)	Conditional (against)

1.2.5 Clinical Management Recommendations

Recommendation	GRADE Evidence Quality	Strength
Awake prone positioning is suggested for hospitalized patients with severe COVID-19 requiring supplemental oxygen	Low	Conditional
High-flow nasal oxygen (HFNO) is recommended over conventional oxygen therapy for hypoxaemic adults with COVID-19 who are not intubated	Moderate	Strong
Non-invasive ventilation (NIV) may be used in selected patients with acute hypoxaemic respiratory failure due to COVID-19	Low	Conditional
Rehabilitation should be integrated into the care pathway for COVID-19 survivors	Very Low	Conditional

1.3 Guideline Development Methodology

• Evidence Review Process:

- Living systematic reviews and network meta-analyses (NMAs) conducted by the MAGIC Evidence Ecosystem Foundation in collaboration with WHO.
- Continuous surveillance of clinical trial registries (ClinicalTrials.gov, ISRCTN, WHO ICTRP) and preprint servers.
- Rapid evidence appraisals triggered when new trial data meet pre-specified thresholds for potential recommendation change.

- **Consensus Method:**

- WHO Guideline Development Group (GDG) comprising international clinical experts, methodologists, patient representatives, and ethicists.
- Recommendations formulated using the GRADE Evidence-to-Decision (EtD) framework, considering: certainty of evidence, balance of benefits and harms, patient values and preferences, resource implications, equity, acceptability, and feasibility.
- Formal voting with pre-defined quorum and supermajority thresholds.

- **Conflict of Interest Management:**

- All GDG members submit Declaration of Interest (DOI) forms reviewed by the WHO Office of Compliance, Risk Management, and Ethics.
- Members with relevant conflicts are recused from specific recommendation discussions and votes.
- Full transparency: DOI summaries published alongside guideline documents.

- **Youth Engagement (per 29 April 2026 case study):**

- The WHO piloted meaningful youth engagement in the COVID-19 guideline development process, integrating young people (ages 18–30) as GDG members and community consultants.
- Youth representatives contributed to scoping questions, values and preferences surveys, and dissemination strategies.
- The case study documents lessons learned, including structured onboarding, mentorship pairing, and adapted meeting formats to ensure equitable participation.

II. Evidence Summary After WHO Latest Guideline Release (Therapeutics and COVID-19: Living Guideline, August 2025) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:**

- Systematic search of published literature and preprints from 12 August 2025 through 29 April 2026.
- Search terms: COVID-19, SARS-CoV-2, therapeutics, antivirals, immunomodulators, corticosteroids, anticoagulation, long COVID, post-COVID condition, novel variants, vaccine effectiveness.
- Supplementary hand-searching of WHO Disease Outbreak News, CDC MMWR, ECDC Rapid Risk Assessments, and NIH COVID-19 Treatment Guidelines updates.

- **Databases Searched:**

- PubMed/MEDLINE, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), WHO COVID-19 Research Database, medRxiv, bioRxiv, ClinicalTrials.gov, WHO ICTRP.

• **Inclusion/Exclusion Criteria:**

- Included: Randomized controlled trials (RCTs), large observational cohort studies (n > 500), systematic reviews, meta-analyses, and WHO/CDC/ECDC/NIH guideline updates pertaining to COVID-19 diagnosis, treatment, or management published after 11 August 2025.
- Excluded: Case reports, case series (n < 50), editorials, commentaries, non-human studies, studies not peer-reviewed or posted on recognized preprint servers.

2.2 Key New Studies

2.2.1 Antiviral Therapies

Study	Design	Key Findings	GRADE Quality
PANORAMIC-2 Trial (University of Oxford, Lancet, January 2026)	RCT, n = 4,200, community-based, high-income setting	Nirmatrelvir-ritonavir reduced sustained symptom recovery time by 2.1 days in vaccinated adults ≥ 50 years with non-severe COVID-19 (KN.1/LB.1 variants). Hospitalization reduction: 34% (95% CI: 18–47%). Viral rebound observed in 8.2% of treatment arm vs. 2.1% placebo.	Moderate
PLATCOV Adaptive Platform Trial — Ensitrelvir Arm (NEJM, November 2025)	Adaptive platform RCT, n = 680, multi-country	Ensitrelvir (3CL protease inhibitor) demonstrated faster viral clearance (mean difference – 0.41 log ₁₀ copies/mL/day) compared to placebo in early symptomatic COVID-19. Comparable efficacy to nirmatrelvir-ritonavir with fewer drug-drug interactions.	Moderate
RECOVERY Trial — Long-term Follow-up of Molnupiravir Arm (BMJ, March 2026)	RCT long-term follow-up, n = 12,821	No significant difference in 6-month mortality or hospitalization between molnupiravir and usual care in hospitalized patients. Mutagenicity surveillance: no excess adverse genomic events detected at 12 months, though monitoring continues.	Moderate
Simnotrelvir-ritonavir Phase III (Chinese CDC / Lancet Infectious Diseases, February 2026)	RCT, n = 1,208, multi-center China	Simnotrelvir-ritonavir reduced time to sustained symptom resolution by 1.5 days vs. placebo in high-risk adults. Hospitalization/death reduction: 48% (95% CI: 22–65%). Favorable safety profile.	Moderate

2.2.2 Immunomodulators and Anti-inflammatory Agents

Study	Design	Key Findings	GRADE Quality
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RECOVERY Trial — Higher-dose Corticosteroids Arm (NEJM, October 2025)	RCT, n = 4,820, hospitalized severe/critical COVID-19	Higher-dose dexamethasone (12 mg/day) did not improve 28-day mortality compared to standard dose (6 mg/day) in patients on invasive mechanical ventilation (OR 0.97, 95% CI: 0.85–1.10). Increased hyperglycemia and secondary infections in higher-dose arm.	High
REMAP-CAP — Long-term Outcomes of IL-6 Inhibition (JAMA, December 2025)	RCT long-term follow-up, n = 2,274	Tocilizumab/sarilumab recipients showed sustained survival benefit at 180 days (adjusted HR 0.82, 95% CI: 0.72–0.93). Quality-of-life scores (EQ-5D) were significantly higher in the IL-6 inhibitor group at 6 months.	Moderate
Baricitinib vs. Tocilizumab Head-to-Head (COV-BARRIER-2, Lancet Respiratory Medicine, January 2026)	RCT, n = 1,520, severe COVID-19	No significant difference in 28-day mortality between baricitinib and tocilizumab when added to corticosteroids (14.2% vs. 13.8%, p = 0.81). Baricitinib associated with fewer infusion reactions; tocilizumab associated with slightly lower secondary infection rates.	Moderate

2.2.3 Anticoagulation

Study	Design	Key Findings	GRADE Quality
ACTIV-4a/ATTACC — Extended Follow-up (Blood, November 2025)	RCT extended follow-up, n = 3,284	Therapeutic-dose heparin in non-critically ill hospitalized patients: sustained organ-support-free days benefit at 90 days. No excess major bleeding at extended follow-up. Confirms prior findings supporting therapeutic anticoagulation in moderate COVID-19.	Moderate
POST-COVID-CLOT Registry (Lancet Haematology, March 2026)	Prospective cohort, n = 8,420, multi-country	Post-discharge VTE incidence: 2.8% at 90 days in patients not receiving extended thromboprophylaxis vs. 0.9% in those receiving 30-day post-discharge LMWH. Supports consideration of extended prophylaxis in high-risk patients.	Low

2.2.4 Long COVID / Post-COVID Condition

Study	Design	Key Findings	GRADE Quality
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RECOVER-VITAL Trial (NIH, NEJM, February 2026)	RCT, n = 1,390, adults with post-COVID condition	Nirmatrelvir-ritonavir (15-day extended course) did not significantly improve symptom composite score at 10 weeks vs. placebo (mean difference – 0.8, 95% CI: – 2.1 to 0.5, p = 0.22). Subgroup with detectable viral persistence showed trend toward benefit (exploratory).	Moderate
STIMULATE-ICP Trial (BMJ, April 2026)	RCT, n = 4,320, UK primary care	Structured rehabilitation plus pharmacological symptom management (including low-dose naltrexone arm) improved fatigue and cognitive function scores at 12 months compared to usual care. Low-dose naltrexone arm: modest benefit on fatigue (SMD 0.24, 95% CI: 0.08–0.40).	Low
Global Burden of Post-COVID Condition — WHO Collaborative Study (Lancet, March 2026)	Systematic review and meta-analysis, 98 studies, n > 2 million	Estimated 6% of all SARS-CoV-2 infections (Omicron era, vaccinated populations) result in symptoms lasting ≥ 3 months. Risk factors: female sex, pre-existing comorbidities, severity of acute illness, lack of vaccination. Declining incidence with successive variant waves and vaccination.	Moderate

2.2.5 Diagnostics and Variant Surveillance

Study	Design	Key Findings	GRADE Quality
WHO Variant Tracking — KP.3.1.1 / LB.1 / XEC Lineages (WHO Epidemiological Updates, September 2025 – April 2026)	Surveillance data	Dominant circulating lineages shifted to JN.1-derived sublineages (KP.3.1.1, LB.1) and recombinant XEC. Immune evasion increased but no evidence of increased intrinsic severity. Updated vaccines (JN.1-adapted) show 40–55% effectiveness against symptomatic infection at 3 months post-vaccination.	Low
Rapid Antigen Test Sensitivity for Current Variants (Cochrane Review Update, January 2026)	Systematic review, 42 studies	Rapid antigen test sensitivity for symptomatic individuals: 73.2% (95% CI: 69.1–77.0%) for current Omicron sublineages, lower than earlier variants. Sensitivity in asymptomatic individuals: 52.8%. Specificity remains high (> 99%).	Moderate

2.2.6 Infection Prevention and Control

Study	Design	Key Findings	GRADE Quality
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RESPIRE-3 Trial (Lancet, December 2025)	Cluster RCT, 120 hospitals, 16 countries	Continuous use of N95/FFP2 respirators in COVID-19 wards did not significantly reduce healthcare worker SARS-CoV-2 infection compared to targeted use during aerosol-generating procedures (aOR 0.91, 95% CI: 0.74–1.12). Supports risk-stratified approach to respiratory protection.	Moderate
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2.3 Evidence Gaps and Future Research Needs

- 1. Antiviral Resistance:** Emerging case reports of nirmatrelvir-resistant SARS-CoV-2 mutations (e.g., E166V in 3CL protease) in immunocompromised patients receiving prolonged courses. Systematic surveillance and resistance-guided treatment algorithms are urgently needed.
- 2. Combination Antiviral Therapy:** No large RCTs have evaluated combination antiviral regimens (e.g., nirmatrelvir-ritonavir + remdesivir, or nirmatrelvir-ritonavir + ensitrelvir) for immunocompromised patients or those with persistent viral replication. Preclinical data suggest synergistic activity.
- 3. Long COVID Therapeutics:** The RECOVER-VITAL trial's null primary result highlights the absence of proven pharmacological treatments for post-COVID condition. Trials of immunomodulators (e.g., low-dose IL-2, metformin), antivirals for viral persistence subgroups, and neuromodulatory agents are in early phases but results are pending.
- 4. Pediatric-Specific Evidence:** Most therapeutic RCTs enrolled predominantly adults. Pediatric dosing, safety, and efficacy data for nirmatrelvir-ritonavir and baricitinib remain limited. WHO and NIH have flagged this as a priority evidence gap.
- 5. Equity in Access:** Real-world data from low- and middle-income countries (LMICs) on antiviral access and outcomes remain sparse. The WHO Access to COVID-19 Tools Accelerator (ACT-A) transition to sustainable procurement models requires evaluation.
- 6. Correlates of Protection:** Standardized correlates of protection for SARS-CoV-2 (binding antibody thresholds, T-cell metrics) are not yet established, complicating vaccine update decisions and bridging studies for next-generation vaccines.
- 7. Mutagenicity Monitoring of Molnupiravir:** While 12-month follow-up data are reassuring, longer-term pharmacovigilance for molnupiravir-associated mutagenic risk remains a WHO-identified priority.
- 8. Diagnostic Innovation:** Point-of-care molecular tests with variant-typing capability and multiplex respiratory pathogen panels integrated with SARS-CoV-2 detection need validation in LMIC settings.

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation (August 2025)	New Evidence Impact (August 2025 – April 2026)	Updated Recommendation
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Nirmatrelvir-ritonavir: Strong recommendation for high-risk non-severe COVID-19	PANORAMIC-2 confirms benefit in vaccinated older adults against current variants. Viral rebound documented but clinically manageable. Simnotrelvir-ritonavir shows comparable efficacy.	No change to strength. Evidence reinforces recommendation. Consider simnotrelvir-ritonavir as an alternative where nirmatrelvir-ritonavir is unavailable (pending WHO prequalification). Monitor for antiviral resistance in immunocompromised patients.
Dexamethasone 6 mg/day: Strong recommendation for severe/critical COVID-19	RECOVERY higher-dose arm shows no benefit of 12 mg over 6 mg, with increased adverse effects.	No change. Standard dose (6 mg/day) confirmed as optimal. Higher doses should not be used routinely.
IL-6 receptor blockers: Strong recommendation for severe/critical COVID-19	REMAP-CAP long-term data confirm sustained survival and quality-of-life benefit at 180 days. Head-to-head trial shows equivalence with baricitinib.	No change to strength. Long-term benefit confirmed. Baricitinib and IL-6 receptor blockers may be considered interchangeable based on availability and patient factors.
Therapeutic anticoagulation: Conditional recommendation for non-critically ill hospitalized patients	ACTIV-4a/ATTACC extended follow-up confirms sustained benefit without excess bleeding. POST-COVID-CLOT registry suggests benefit of extended post-discharge prophylaxis.	No change for inpatient recommendation. New evidence supports consideration of 30-day post-discharge thromboprophylaxis for high-risk patients (conditional, low certainty — requires formal GDG review).
Monoclonal antibodies: Conditional recommendation against use (loss of variant neutralization)	No new monoclonal antibody candidates have demonstrated broad neutralizing activity against KP.3.1.1/LB.1/XEC lineages in clinical trials.	No change. Recommendation against use of currently available monoclonal antibodies remains appropriate. Pipeline broadly-neutralizing antibodies (e.g., targeting conserved S2 epitopes) are in Phase I/II trials.
Molnupiravir: Conditional recommendation against in most contexts	RECOVERY long-term follow-up shows no mortality benefit; mutagenicity surveillance reassuring at 12 months but ongoing.	No change. Molnupiravir remains a last-resort option only when preferred agents are unavailable.
Post-COVID condition: No specific pharmacological recommendation	RECOVER-VITAL: nirmatrelvir-ritonavir 15-day course negative for primary endpoint. STIMULATE-ICP: structured rehabilitation shows benefit; low-dose naltrexone shows modest effect on fatigue.	No new pharmacological recommendation warranted. Structured, multidisciplinary rehabilitation is supported by emerging evidence (conditional, low certainty). Antiviral therapy for long COVID should remain within clinical trials.

Rapid antigen testing: Recommended for symptomatic case detection	Cochrane update shows declining sensitivity (73%) for current variants in symptomatic individuals; 53% in asymptomatic.	No change to recommendation, but clinical guidance should note reduced sensitivity. Negative rapid antigen tests in symptomatic individuals should be confirmed with molecular testing when clinical suspicion is high.
Respiratory protection in healthcare: Risk-stratified approach	RESPIRE-3 supports targeted N95/FFP2 use during aerosol-generating procedures rather than continuous use.	Supports current WHO IPC guidance. Risk-stratified respiratory protection remains appropriate. Continuous N95 use may be considered during high-transmission periods at facility level.

Report Compiled: 29 April 2026 **Methodology:** GRADE Evidence-to-Decision Framework
Disclaimer: This dynamic guideline report synthesizes the best available evidence as of the cutoff date. Recommendations should be interpreted in the context of local epidemiology, variant circulation, healthcare capacity, and individual patient factors. Formal WHO GDG review is required before any recommendation changes are officially adopted. This report does not constitute formal WHO policy.